PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis
AUTHORS	Burns, Alex; Donnelly, Brian; Feyi-Waboso, Joshua; Shephard, Elizabeth; Calitri, Raff; Tarrant, Mark; Dean, Sarah

VERSION 1 – REVIEW

REVIEWER	Nadeem Qureshi University of Nottingham, of Primary Care
REVIEW RETURNED	24-Jan-2022

GENERAL COMMENTS	This is a highly relevant study of risk assessment tools in primary care. It is highly applicable to the current approaches to clinical management Considering each section Abstract An impressive search strategy and number of studies reviewed is documented Major amendments In the results it is not clear what are the key findings from the qualitative analysis. Could do with more detailed (appreciating the limited word count) Minor amendments It would be helpful to have a clearer conclusion from the results, for example, specifically how risk assessment tools influenced the consultation Introduction Well-written introduction setting the scene for the use of electronic risk assessment tools. Major amendments Suggest rephrasing the AIMS, as it reads it is not very clear, for example, are authors' saying: assess how risk assessment tools affects communication and understanding of disease risk when used to make diagnoses

	Methods Clear presentation of the analytic methods is provided by the authors Major amendments It may be worth adding if PICo (Popolation, Phenomena of interest, Context) was used to develop the study search strategy. I notice authors describe Systm One but there are also risk assessment tools used in other GP IT systems, such as, EMIS. Is it worth mentioning this as well? Results The authors provide comprehensive details about the included qualitative studies. Also they have included interesting concepts, for example, description of “Fear” Major amendments: Initially in paper the authors describe differential diagnosis that use different terms in the results, such as, page 6, line 7 “potential possible diagnosis”, then in line 54 “alternative diagnoses”. Would be helpful to be consistent. The first line of the results (line 3) about meta-study may be better placed in the method section I notice that the authors introduce findings about risk status, such as stating novel risk on line 45, page 6 . Also they describe best practice on line 14, page 7. Referring back to the aims of the study, discussion of risk and best practice do not seem to fit directly with those currently presented. This may simply need rephrasing of the aims and the rationale for this systematic review in the introduction. Of course if the best practice relates to best practice for diagnosis, this still falls within the current aims as written. Similarly the line 11 on page 7, the authors describe missing information. This fits with the aims if related to missing information to help the diagnosis, but if this relates to other missing information then they should explain what the missing information is about and should fit with the aims of the study. Line 50 on page 7, the comment about "how clinicians normally think about the diagnosis". Did these comments come from the clinician and/or patients? On page 8, line 20, I'm unclear what the authors mean when describing "preference from both clinicians and patients were diagnostic tests" Table 1 is a useful presentation of the structure of the included studies. Minor amendments: Table 3 would this be better presented as a supplementary table. Discussion

	The authors make salient points about the strengths of this systematic review. Also noting underrepresentation of patients regarding risk assessment tools is highly relevant. There was also a nice presentation of the interpretation of fear in the diagnosis. Major amendments The summary section (possibly better described as main findings) was rather confusing to read. This may be better rephrased to present how the themes link to the risk assessment tools role and impact, with reference to other studies moved to the “comparison with other literature” section Line 49 - 50 on page 8, I'm not clear what is meant by conceptual design in in this line and how these findings are presented in the results Line 23 on page 9, why is translation in quotation marks and I'm not clear what this means. Line 35 on page 9, how is "gut feelings" presented in the results? Also it would be helpful to have a paragraph about future research recommendations. I would suggest this is a highly relevant study which should be published but does need some significant amendments before it is suitable for publication.

REVIEWER	Christoffer Bjerre Haase University of Copenhagen, Dept. of Public Health
REVIEW RETURNED	25-Jan-2022

GENERAL COMMENTS	Thank you very much for this great and interesting paper. It is an important topic and the paper is overall great, clear, and thorough. See my comments and questions below. Please consider them more as my suggestions or thoughts instead of criticism as I really enjoyed the manuscript. Regarding the aim, main concepts, and introduction: I am not sure that I fully understand the stated aim or purpose of the study. You state that the aim is to “examining how they affect the communication and understanding of diagnostic risk and uncertainty” (my underline). Yet, later you explain that “This review examines how eRATs change the perception of risk and uncertainty”. Later, “the purpose of this review is to examine how clinicians and patients interact with eRATs”. While ‘interaction’ may refer to ‘communication and understanding’, I am not sure that it refers to ‘change the perception’. I also think that ‘communication and understanding’ and ‘change the perception’ are two different things. In addition, since it is a qualitative study, I would prefer if the introduction included some lines about your theoretical position regarding these key concepts that you are investigating: How do you understand communication and understanding (if these are the main concepts). If it is about changing perception, which theoretical background are you using to understand this in relation to eRATs?

	As I know the word count is limited, I think you could reduce and combine paragraph two (line 13: “Clinical Decision Support Systems are designed to…”) and paragraph three (line 23: “eRATs can be considered as the computerized evolution of clinical scoring systems”…). I think the introduction is well written and eRATs are well explained but in order to get the most out of the findings in the results, I think the readers need to know more about the main concepts that are used to understand eRATs. Regarding method: To the best of my knowledge, you have successfully conducted a transparent and exhaustive search for relevant studies, assessed the risk of bias of the primary studies, and stated the overall methods for the synthesis of qualitative research. It is well written, very clear and so are the tables and figures. Regarding results: As it is a qualitative study, and in order to get most out of the findings, I would prefer less quantitative findings and more qualitative analysis. Except for the publication dates, I would prefer removing the paragraphs from page six, line 13 to line 29 or reduce it to the most important. I am curious if you could elaborate on the statement “It became evident that there were three conceptually different forms of eRATs:…”. Why was this temporal concept evident? Could other concepts be used to differentiate the eRAT? Spatial or social concepts or simply differences regarding actors, conditions/diagnoses, accountability, etc.? Playing with different ways to order the differences could give some interesting or inspiring insights about which aspects were in focus and which aspects were taken for granted. Secondly, I am not sure why you categorize the eRATs into these three forms, as they are not used later in the results or discussion. Could you elaborate on the importance of this distinction? I would prefer more analysis in the analytical themes. What differences/extremes/tensions/conflicts/contradictions were in play in each of the themes? Also, for each theme, I am surprised that no distinctions are made between GPs and patients. Did GPs and patients always share the same perception of risks, autonomy, communication and mistrust? It would really be interesting to read some analysis about the differences between them in each theme. Regarding Discussion: After reading the result, I am not fully convinced about the first claim: “This review found that eRATs can be helpful in considering and communicating risk and uncertainty during diagnosis”. I am not sure for whom and how it can be helpful and on expenses of what. Is it for example helpful for the GP to consider and communicate a risk that the GP had never thought relevant? And how is it helpful if there was a lack of trust in the risk output of the eRAT? Claims like this indicate that the paper could benefit from a more thorough analysis that includes more elaboration of the main concepts and the agents/subjects that are analyzed. I am not sure that I understand this statement: “This review examines how eRATs change the perception of risk and uncertainty: it could be argued that when the language surrounding an eRAT fails to describe it in such terms (i.e. “electronic” or “computerised”), it is no longer a challenge to the way diagnoses are considered, because it is now the routine”. Do you argue for it or not? And why is

	that? Could you be more specific what you mean with a “challenge to the way diagnoses are considered”? And is it not possible that it is still a challenge even though it has become a routine? The statement that “There was an underrepresentation of patient’s views regarding eRATs” sounds too quantitative to me. I think it depends on the analysis and the aim. I would also suggest reducing the paragraph from page eight, line 55 to page nine, line 12. I agree that the changes of technology terminology is challenging. Yet, this can be stated shortly. Finally, just a question: did the concept of accountability not appear in the literature? Personally, I think this is an interesting finding. Thank you again for this great manuscript.

REVIEWER	Mary Dahm Australian National University, Institute for Communication in Healthcare
REVIEW RETURNED	08-Feb-2022

GENERAL COMMENTS	A very timely review and well-written paper. What I feel is lacking is a more thorough presentation of results in relation to what the authors promise in other sections and an expansion of the discussion section. This could be easily improved and I make some suggestions below. There are several issues in the results section that need attention 1. The authors described that "three conceptually different forms of eRATs" were identified and that authors wanted "to explore the themes across these eRATs, to see if there were findings consistent between them." However, in the subsequent section on Analytical themes, it is not clear which theme(s) related to which (or all) of the eRAT types. This is even more problematic in relation to the discussion stating "Some themes presented derive a greater part of their data from eRATs of a particular conceptual design, which could challenge their transferability across all eRATs" I would suggest that it should be made clearer in the results section, which type of eRAT was associated with which theme. 2. While important the discussion around the nomenclature of tests introduces new data that should have been presented earlier in the result section, as this is currently the first time these different names are mentioned. 3. There are also reference to subsets of studies (e.g. "perception and communication of risk and uncertainty was included in the analytical findings of only ten of the primary studies, limiting this aspect of the synthesis"). It is not immediately clear where in the spectrum of eRATs these studies sit. Extending the table of included studies, including type of eRAT, nomenclature, and relevant themes associated with the study would help solve the issues addressed above. Quotes in the manuscript, as well as supplementary material, could also reference the type of eRAT, if feasible. Table1 which I assume is part of the main manuscript rather than supplementary material, is not referred to in the manuscript at all. The discussion of 'gut feelings' would benefit from a brief addition, in relation to challenging gut feelings to overcome cognitive biases,

	eRATs can become a trigger for debiasing as described under theme "novel risk". Suggest including ref to e.g.: https://doi.org/10.1097/00001888-200308000-00003 https://doi.org/10.1111/medu.12670 https://doi.org/10.1136/bmjqs-2012-001712 https://doi.org/10.1016/j.pec.2021.05.012 A further brief discussion of results in relation to "diagnostic uncertainty" literature would further strengthen this section. Suggest to incorporate e.g.: https://doi.org/10.1016/j.pec.2019.06.012 https://doi.org/10.1007/s11606-017-4164-1 The authors include an ENTREQ checklist but not a PRISMA checklist (which is the one I would have expected but might be wrong given the second qualitative analytical approach embedded in the systematic review). Should it have both? OTHER Some minor typos/ missing/ wrong punctuation throughout " eRATs12,13, There have been no systematic reviews using qualitative methods examining risk and uncertainty in their use. " "The first is the fear of a serious diagnosis when the eRAT triggers {WHAT?}."

VERSION 1 – AUTHOR RESPONSE

Reviewer 1			
In the results it is not clear what are the key findings from the qualitative analysis. Could do with more detailed (appreciating the limited word count) It would be helpful to have a clearer conclusion from the results, for example, specifically how risk assessment tools influenced the consultation	8	Many thanks. I have reworded the “summary” part of the discussion to make this clearer.	See above (Discussion Summary in paper)
Suggest rephrasing the AIMS, as it reads it is not very clear, for example, are authors' saying: assess how risk assessment tools affects communication and understanding of disease risk when used to make diagnoses	3	Many thanks. Your suggestion reads well and captures the aim. I have reworded it accordingly.	This study aims to synthesise qualitative research on eRATs in the primary care consultation, examining how they affect the communication and understanding of the risk of particular disease, and uncertainty when used

			making diagnoses.
It may be worth adding if *PICo (Popolation, Phenomena of interest, Context)* was used to develop the study search strategy.	4	The search terms have been reordered to conform to the PICO structure.	The search strategy[14] targeted studies involving a primary care population, where risk assessment or decision support tools are used for, where qualitative data are reported regarding risk or uncertainty.
I notice authors describe Systm One but there are also risk assessment tools used in other GP IT systems, such as, EMIS. Is it worth mentioning this as well?	4	The use of Systm One was for practicality. There was no immediate access to an EMIS system in the study team. Risk assessment tools are extensive, but evidence for their use is scant: the look at System One was a practical and pragmatic method of ensuring the "main" ones in use were being covered (if evidence existed) in this review. It was not intended to be a comprehensive search. System One also has an optional extra functionality in the form of "Ardens" templates. These contain many more risk assessment tools, but were not covered for	

		practical time reasons also.	
Initially in paper the authors describe differential diagnosis that use different terms in the results, such as, page 6, line 7 “potential possible diagnosis”, then in line 54 “alternative diagnoses”. Would be helpful to be consistent.	5&8	Many thanks for spotting this. I have changed this so it is consistent.	
The first line of the results (line 3) about meta-study may be better placed in the method section	5	I have reworded this section, which has removed the mention.	
notice that the authors introduce findings about risk status, such as stating novel risk on line 45, page 6 . Also they describe best practice on line 14, page 7. Referring back to the aims of the study, discussion of risk and best practice do not seem to fit directly with those currently presented. This may simply need rephrasing of the aims and the rationale for this systematic review in the introduction. Of course if the best practice relates to best practice for diagnosis, this still falls within the current aims as written.	n/a	Yes. On my reading again, the finding concerning education best practice does not fall into the remit of the aims of the study. I will remove the line regarding best practice. Many thanks for this spot.	
Similarly the line 11 on page 7, the authors describe missing information. This fits with the aims if related to missing information to help the diagnosis, but if this relates to other missing information then they should explain what the missing information is about and should fit with the aims of the study.	6	This was not expressed this well. Thanks for pointing this out: it is missing information relating to the diagnosis. I have reworded this section.	triggering them to add missing information to the eRAT,[37] suggesting the next diagnostic tests which are appropriate[24] to reduce uncertainty and improve diagnostic accuracy.
Line 50 on page 7, the comment about "how clinicians normally think about the diagnosis". Did these comments come from the clinician and/or patients?	7	Many thanks for pointing out the ambiguity. I have reworded to clarify.	A second fear from clinicians, was of the eRAT interrupting how they normally think about

			diagnosis
On page 8, line 20, I'm unclear what the authors mean when describing "preference from both clinicians and patients were diagnostic tests"	7	Thanks for this. I have altered the wording.	Both clinicians and patients preferred diagnostic tests over the output of an eRAT. Here, a diagnostic test would be ordered despite the eRAT suggesting a risk low enough that testing was unnecessary: the test was trusted and the eRAT discounted
Table 3 would this be better presented as a supplementary table.	n/a	I will add as a supplementary table.	
The summary section (possibly better described as main findings) was rather confusing to read. This may be better rephrased to present how the themes link to the risk assessment tools role and impact, with reference to other studies moved to the “comparison with other literature” section	8	Thanks for this advice. I have reworded the summary. However, the subheadings are those of the BMJ Open editorial policy.	See above (Discussion Summary in paper)
Line 49 - 50 on page 8, I'm not clear what is meant by conceptual design in in this line and how these findings are presented in the results	8	It refers to the section on the different forms of eRAT that became evident, as described in the results section. This was not clear to this reviewer, so I have explicitly linked it in the text.	Some themes presented derive a greater part of their data from eRATs of a particular conceptual design (“clinical”, “prognostic” or “differential diagnosis”, as described in the results section), which could challenge their transferability across all eRATs.
Line 23 on page 9, why is translation in quotation	8	Many thanks. Translation is in	Thematic synthesis includes

marks and I'm not clear what this means.		quotation marks as it is used in a non-literal sense here, referring as it does to the process described in thematic synthesis. I have tried to clarify this with a quote from the original methodological citation.	"translation" of the review findings into the primary studies' thematic frameworks. This involves "*the process of taking concepts from one study and recognising the same concepts in another study*".
Line 35 on page 9, how is "gut feelings" presented in the results?	9	I have added a line to try and make this flow more logically to gut feelings.	Part of clinical judgement is a clinicians' gut feelings
Also it would be helpful to have a paragraph about future research recommendations	9	I have added this to the "implications" paragraph at the end.	Future research is needed into the understanding of diagnostic risk and uncertainty, from both patient and clinician's view points, how they differ and how eRATs influence the resulting communication. More evidence is needed examining how eRATs influence a clinicians clinical reasoning, particularly when the eRAT challenges clinical judgement.
Reviewer 2			
I am not sure that I fully understand the stated aim or purpose of the study. You state that the aim is to "examining how they affect the communication and understanding of diagnostic risk and uncertainty" (my underline). Yet, later you explain that "This review examines how eRATs change the perception of risk and uncertainty". Later, "the purpose of this review is	3, 8	Many thanks for pointing out the inconsistency in wording: I used "perception" when a better word was "understanding".	This study aims to synthesise qualitative research on eRATs in the primary care consultation,

to examine how clinicians and patients interact with eRATs". While 'interaction' may refer to 'communication and understanding', I am not sure that it refers to 'change the perception'. I also think that 'communication and understanding' and 'change the perception' are two different things.		This has been replaced. The word interaction was used as a shorthand as well. I have added a clarifier.	examining how they affect the communication and understanding of the risk of a particular disease, and uncertainty when used making diagnoses. This review is the first to address how eRATs affect understanding and communication of risk and uncertainty in the primary care consultation. This review was intended to look at the wider clinician and patient interaction (in terms of understanding and communication)
I would prefer if the introduction included some lines about your theoretical position regarding these key concepts that you are investigating: How do you understand communication and understanding (if these are the main concepts). If it is about changing perception, which theoretical background are you using to understand this in relation to eRATs?	3	Many thanks for this. I have added a paragraph regarding our theoretical position.	Diagnosis in primary care tends to be the outcome of clinical reasoning, which can be considered using the theories of Situativity:[11] that diagnosis will depend on other actors (as well as the clinician) present during the consultation, and

			their interaction with multiple environmental inputs, in our case eRATs. Within this clinical situation we are interested in the communication of risk, which is the best proxy we have of the understanding of risk. The Theory of Communicative Rationality [12] may help our understanding - communication can be emancipative: an attempt to achieve mutual understanding (as in shared decision making); or strategic, where the intention of the communication is to achieve a particular aim of an individual, such as calculative manipulation of others.
As I know the word count is limited, I think you could reduce and combine paragraph two (line 13: "Clinical Decision Support Systems are designed to…") and paragraph three (line 23: "eRATs can be considered as the computerized evolution of clinical scoring systems"…). I think the introduction is well written and eRATs are well explained but in order to get the most out of the findings in the results, I think the readers need to know more about the main concepts that are used to understand eRATs.	3	I have reduced the paragraph in question	Clinical Decision Support Systems are designed to "enhance medical decisions with targeted clinical knowledge, patient information, and other health information"[5]. "Electronic Risk Assessment

			Tools" (hereafter referred to as eRATs) are decision support systems which help with diagnosis. They are software where information about a patient is processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future. The output can be a quantitative risk of a specific diagnosis, or a more qualitative output: prompting the clinician to consider a particular diagnosis or list of possible diagnoses. eRATs can enable extraction of data previously inputted into primary care IT systems, and can prompt users to input missing data. They can incorporate "pop ups" during a consultation to avoid missing important diagnoses.[6,7] They have a range of stated aims: making diagnosis more evidence based and objective; avoiding missed diagnoses; or

			facilitating shared decision making.
As it is a qualitative study, and in order to get most out of the findings, I would prefer less quantitative findings and more qualitative analysis. Except for the publication dates, I would prefer removing the paragraphs from page six, line 13 to line 29 or reduce it to the most important.	5	It is a systematic review and qualitative synthesis. The reporting of the studies is to demonstrate the range of studies found in the systematic review, to validate the subsequent qualitative synthesis. I acknowledge that there is a table which reports these findings though and have reduced the paragraph in question	Included studies had publication dates ranging from 2003 to 2019.[21,23–49] There was a broad range of targeted diagnoses, as well as data collection and analytical methods. There was also a range of nomenclature used to describe the eRAT being used. Details are available in Table 1. Only ten of the studies reported themes or findings pertaining to risk or uncertainty in their analysis
I am curious if you could elaborate on the statement "It became evident that there were three conceptually different forms of eRATs:…". Why was this temporal concept evident? Could other concepts be used to differentiate the eRAT? Spatial or social concepts or simply differences regarding actors, conditions/diagnoses, accountability, etc.? Playing with different ways to order the differences could give some interesting or inspiring insights about which aspects were in focus and which aspects were taken for granted. Secondly, I am not sure why you categorize the eRATs into these three forms, as they are not used later in the results or discussion. Could you elaborate on the importance of this distinction? I would prefer more analysis in the analytical themes. What differences/extremes/tensions/conflicts/contradictions were in play in each of the themes?	5	Thank you for highlighting this. I have tried to encapsulate the primacy of this concept and added to this part of the paper.	Three conceptually different forms of eRATs were identified: those which produced a risk of an immediate, or "clinical" diagnosis, (e.g., the risk of a current bacterial tonsillitis); those which produced a risk of a future or "prognostic" diagnosis, (e.g., the risk of cardiovascular disease in the next decade); and those which produced a list of

			potential alternative diagnoses, or "differential diagnosis generators". This is an important distinction because it gets to the heart of what diagnosis is: separating the affected from the non-affected. This means defining illness, and the concept of current illness is distinct to the threat of future illness.[20]
Also, for each theme, I am surprised that no distinctions are made between GPs and patients. Did GPs and patients always share the same perception of risks, autonomy, communication and mistrust? It would really be interesting to read some analysis about the differences between them in each theme.	5-7	Previous drafts of the paper had more regarding this, and I have reinserted/reedited the analysis.	See Analytical Themes in results section
After reading the result, I am not fully convinced about the first claim: "This review found that eRATs can be helpful in considering and communicating risk and uncertainty during diagnosis". I am not sure for whom and how it can be helpful and on expenses of what. Is it for example helpful for the GP to consider and communicate a risk that the GP had never thought relevant? And how is it helpful if there was a lack of trust in the risk output of the eRAT? Claims like this indicate that the paper could benefit from a more thorough analysis that includes more elaboration of the main concepts and the agents/subjects that are analyzed.	8	Many thanks. It was intended to read more measured and nuanced. I have reworded this section.	See above (Discussion Summary in paper)
I am not sure that I understand this statement: "This review examines how eRATs change the perception of risk and uncertainty: it could be argued that when the language surrounding an eRAT fails to describe it in such terms (i.e. "electronic" or "computerised"), it is no longer a challenge to the way diagnoses are considered, because it is now the routine". Do you argue for it or not? And why is that? Could you be more specific what you mean with a "challenge to the	n/a	Many thanks for this. We have reconsidered this issue and realise it is not central to our discussion. The statement has been removed.	

way diagnoses are considered"? And is it not possible that it is still a challenge even though it has become a routine?			
The statement that "There was an underrepresentation of patient's views regarding eRATs" sounds too quantitative to me. I think it depends on the analysis and the aim. I would also suggest reducing the paragraph from page eight, line 55 to page nine, line 12. I agree that the changes of technology terminology is challenging. Yet, this can be stated shortly.	8	Although there was quantitatively less data, it was also less "rich", and I have reworded this, and have shortened the paragraph in question regarding terminology.	There was limited data from patient's views regarding eRATs: the majority of the qualitative data regarding risk and uncertainty came from clinicians in the review's studies, and data from patients was less conceptually rich.
Finally, just a question: did the concept of accountability not appear in the literature? Personally, I think this is an interesting finding.	n/a	Thank you for the question. In the process of analysis, accountability was not specifically mentioned, although it was alluded to. The concept of "fear" included a clinicians fear of accountability for a patients outcome, which would be increased if they went against an eRAT output.	
Reviewer 3			
The authors described that "three conceptually different forms of eRATs" were identified and that authors wanted "to explore the themes across these eRATs, to see if there were findings consistent between them." However, in the subsequent section on Analytical themes, it is not clear which theme(s) related to which (or all) of the eRAT types. This is even more problematic in relation to the discussion stating "Some themes presented derive a greater part of their data from eRATs of a particular conceptual design, which could challenge their transferability across all eRATs" I would suggest that it should be made clearer in the results section,	n/a	I have added the type of eRAT to the parts of the analysis where it is most pertinent. See below re table 1	

which type of eRAT was associated with which theme.			
While important the discussion around the nomenclature of tests introduces new data that should have been presented earlier in the result section, as this is currently the first time these different names are mentioned.	n/a	Added to table 1 see below	
There are also reference to subsets of studies (e.g. "perception and communication of risk and uncertainty was included in the analytical findings of only ten of the primary studies, limiting this aspect of the synthesis"). It is not immediately clear where in the spectrum of eRATs these studies sit.	n/a	Now visible via changed table 1 see below.	
Extending the table of included studies, including type of eRAT, nomenclature, and relevant themes associated with the study would help solve the issues addressed above. Quotes in the manuscript, as well as supplementary material, could also reference the type of eRAT, if feasible.	n/a	Many thanks. This is a helpful suggestion. I have added to table1 the type of eRAT as well as the nomenclature used to describe it in the initial study.	See table 1
The authors include an ENTREQ checklist but not a PRISMA checklist (which is the one I would have expected but might be wrong given the second qualitative analytical approach embedded in the systematic review). Should it have both?	n/a	PRISMA is a checklist designed with quantitative methods in mind. ENTREQ is specifically designed for Qualitative SR, and is on the Equator Network's website for this purpose (BMJ open publishing guidelines stipulate the use of checklists from this website)	
The discussion of 'gut feelings' would benefit from a brief addition, in relation to challenging gut feelings to overcome cognitive biases, eRATs can become a trigger for debiasing as described under theme "novel risk".	8	Many thanks. A comment on the debiasing intention of eRATs has been added to the discussion section	An eRAT introducing "Novel" risk in a consultation could be seen as an example of this, and could have the potential to overcome the

			cognitive errors of misdiagnosis (such as premature closure).
The authors include an ENTREQ checklist but not a PRISMA checklist (which is the one I would have expected but might be wrong given the second qualitative analytical approach embedded in the systematic review). Should it have both?	n/a	PRISMA is a checklist designed with quantitative methods in mind. ENTREQ is specifically designed for Qualitative SR, and is on the Equator Network's website for this purpose (BMJ open publishing guidelines stipulate the use of checklists from this website)	

VERSION 2 – REVIEW

REVIEWER	Christoffer Bjerre Haase University of Copenhagen, Dept. of Public Health
REVIEW RETURNED	19-May-2022

GENERAL COMMENTS	Thank you for allowing me to read this great paper again. You have definitely improved it. I only have a few comments/questions: The conclusion in the abstract: I am not sure that I am convinced that you can conclude this based on your results and discussion. Could you be more specific of what is essential in your findings and the implications of this? You write "disease prevalence" on line 7. You could consider using the term pretest probability instead. In the result on line 28, you write "this is an important distinction because […] what diagnosis is […]". In the literature, people claim diagnosis to be many different things based on many different arguments. But are the three different concepts of eRATS that you describe not important for other, more concrete reasons? Such as, having different implications for GPs and patients, including responsibilities for each of them as well as the producers of the eRATs? For example, a risk of a future condition put more responsibility on the patient (whether the estimated risk is true or not) than the GP compared to an eRAT suggesting a differential diagnosis, which put an acute responsibility on the GP.

	I am sorry, but I do not understand the following sentences: “The data was synthesised data the three conceptual forms, and themes across these eRATs,explored to see if there were findings consistent between them. This acknowledges the aspects of the different eRATs which are the same and contributory to the aims of the review.” – could you rephrase? For the theme ‘communication’, I am not convinced by the two quotes: ““OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that.” – How is it mainly related to communication? Is it not more about ‘second opinion’ or ‘decision support’? Same with ““I think FRAX [is] useful because [it] can cut down on unnecessary investigations” – where it the communication aspect in this? Is it not about reducing work for the GP? For the theme ‘autonomy’, I am not convinced by the quote: “Anything that gets people to have a think about their health is a good thing, anything at all” – what does that have to do with autonomy? Is it not about awareness? Or the opposite of autonomy since people have to think about their health? Also with the theme ‘fear’ with the quote: ““And I thought, ‘Wow! Patients coming with these tummy pains,’ even if I’m thinking, ‘She’s got a 10% chance,’ to actually think, ‘Wow! Listen, you got a 38% chance of cancer.’ General public are very—I would find that confronting and the general public have a shocking record at understanding risk.”” – Is it not about various conceptions of risk? Thank you.

VERSION 2 – AUTHOR RESPONSE

Reviewer 2 comments 25/5/22			
The conclusion in the abstract: I am not sure that I am convinced that you can conclude this based on your results and discussion. Could you be more specific of what is essential in your findings and the implications of this?	P2	Many thanks. We have reworded this section.	eRATs may improve the understanding and communication of risk in the primary care consultation. The themes of “Fear” and “Mistrust” represent potential challenges which eRATs may create.
You write “disease prevalence” on line 7. You could consider using the term pretest probability instead.	P3	Thank you. We agree: it is more precise, and we think I may have put in a previous version, but thought it might be less familiar term for the readership. We have reintroduced the term in the revision	Pre-test probability is lower than other health-care settings.
In the result on line 28, you write “this is an important	P5	Thank you for this insight, which we agree with. There	It has further implications in that both clinician and patient

distinction because […] what diagnosis is […]". In the literature, people claim diagnosis to be many different things based on many different arguments. But are the three different concepts of eRATS that you describe not important for other, more concrete reasons? Such as, having different implications for GPs and patients, including responsibilities for each of them as well as the producers of the eRATs? For example, a risk of a future condition put more responsibility on the patient (whether the estimated risk is true or not) than the GP compared to an eRAT suggesting a differential diagnosis, which put an acute responsibility on the GP.		are also other implications of this difference in eRATs. The risk of a future disease does introduce a possibility of agency, or ability to change the outcome for both clinician and patient (although we agree with your aside about estimated risk, and we do not think decision support tools are good in demonstrating how much risk we have agency over). But it might also be argued that this is a consequence of the difference in type of risk (and thus type of eRAT), rather than integral as to WHY it is different. A comment has been added to this effect.	may be able to change, and perhaps have responsibility for, "prognostic" risk.
I am sorry, but I do not understand the following sentences: "The data was synthesised data the three conceptual forms, and themes across these eRATs,explored to see if there were findings consistent between them. This acknowledges the aspects of the different eRATs which are the same and contributory to the aims of the review." – could you rephrase?	P5	Thank you. This has been reworded.	It was decided that the three conceptual eRATs had aspects which were similar enough to attempt a synthesis: to see if there were findings which applied across the forms of eRATs, and that could help answer the research question.
For the theme 'communication', I am not convinced by the two quotes: ""OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that." – How is it mainly related to communication? Is it not more about 'second opinion' or 'decision support'?	P6	Qualitative data can be read in different ways, and one might read the quotes you mention in the manner you describe. In our analysis though, we went a step further than saying the eRAT was a decision support tool, we felt that eRAT here was also being used as a communicative tool: the outcome of the eRAT being used as a justifier of the next course of action; and the	They can provide a focus point for a consultation.[39,46] Clinicians can use the eRAT output to justify, and explain to the patient, their next action, be this referral or further testing particularly in eRATs for clinical diagnosis.[24,45] Where the risk output is low they can justify inaction.[24,31] The eRAT acts as a tool to communicate this:

Same with ""I think FRAX [is] useful because [it] can cut down on unnecessary investigations" – where it the communication aspect in this? Is it not about reducing work for the GP?		words this clinician would use to explain this to the patient. We have added a line which I hope clarifies this.	
For the theme 'autonomy', I am not convinced by the quote: "Anything that gets people to have a think about their health is a good thing, anything at all" – what does that have to do with autonomy? Is it not about awareness? Or the opposite of autonomy since people have to think about their health?	P6	We agree that this quote could be viewed through the lens of awareness as well as the outcome we came to in our analysis. Autonomy (in medicine at least) is the right of competent adults to make informed decisions about their own care. We took this "thinking about health" to be informing, and thus increasing autonomy. I have added a line which I hope makes this link explicit.	The eRAT was an enabler of this as it provided a personalised risk and prompted patients to think about it.
Also with the theme 'fear' with the quote: ""And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."" – Is it not about various conceptions of risk?	P7	Thank you. Here we are specifically interrogating our data with regard to potential negative experiences: a clinician is worried about "confronting" the patient, who might struggle with understanding the information and we took this to mean that it may produce fear.	

VERSION 3 – REVIEW

REVIEWER	Christoffer Bjerre Haase University of Copenhagen, Dept. of Public Health
REVIEW RETURNED	09-Jun-2022

GENERAL COMMENTS	Thank you for the replies and changes.